

TASK 1

Carrier: At first glance, CARRIER had missing data. However, this was due to pandas interpreting its IATA code 'NA' as missing. As a result, no imputation was necessary and instead I performed a simple lookup, mapping North American Airlines to NA.

Carrier Name: I found that missing rows corresponded to the carrier codes 'L4' and 'OH'. A lookup revealed that the L4 code mapped to Lynx Aviation d/b/a Frontier Airlines so I mapped the missing rows to match that. For the remaining missing rows, I chose not to impute the remaining 97 rows. OH underwent a carrier code reassignment and 2013, the year with all the missing OH values was during this transition.

Airline ID: All 105 missing rows were corresponding to the same rows where CARRIER_NAME was missing. I performed a lookup using non-missing rows to build a lookup table between Airline_ID and Carrier Name and filled in the gaps. The remaining missing AIRLINE_ID rows correspond to the unrecoverable OH transition period rows where CARRIER_NAME was also absent, so no further imputation was possible.

Manufacturer: No imputation was applied. The 3 missing values could not be reliably recovered from other columns in the dataset as this is a MNAR case.

I think that both Capacity in Pounds and Number of Seats are MAR. I graphed each column against other columns in the dataset, which revealed their categories of missingness. Number of Seats is primarily conditional on Aircraft Status as status L, O, and A had much higher missingness rates than A. Capacity in Pounds is primarily conditional on Year as the year 2023 had 100% of its data missing. Additionally, Aircraft Status 'L' and 'O' had 100% missing as well, suggesting it could be a predictor.

I then tested different imputation methods and compared their RMSE values. For Capacity in Pounds, predictive mean matching had the lowest RMSE of 59385.4684 and for Number of Seats, KNN had the lowest RMSE of 8.3563.

TASK 2

Standardization was required for all the columns of interest:

Operating Status - Contained casing inconsistencies. I found that 'Y' and 'y' correspond to the same thing so I standardized it to have capital letters Y and N only. I found one case of an empty string which I changed to NaN since the next step involves dumping remaining missing values.

Aircraft Status - Also contained casing inconsistencies. The lowercase variants appeared to be the same categories as their uppercase counterparts. I standardized all values to uppercase using `.str.strip().str.upper()`, reducing the column to 4 clean categories: 'A', 'B', 'L', 'O'.

Manufacturer -This column had trailing whitespace, mixed casing, concatenated words, typos, and leasing company names recorded in place of actual manufacturers. I first stripped whitespace, then applied a mapping focused on the highest-frequency variants identified through `value_counts()`. For lessor/manufacturer relationships, I grouped them based on the most common among names. For example, 'EASYJET' : ['easyJet/BBAM', 'easyJet/Fuyo', 'easyJet/GECAS', 'easyJet/GOAL', 'easyJet/Investec', 'CBA/EASYJET', 'CBA/easy'. They all have EasyJet in common so I labelled them as one value. Lastly, I made all rows upper case.

Model- Many variations existed for the same aircraft model due to inconsistent prefixes ('737-800' vs 'B737-800' vs 'B-737-800'), passenger/cargo suffixes ('-PSGR', 'PAX', 'PASSENGERONLY'), and varying levels of specificity in variant codes. I used `value_counts()` to identify the most frequent model families and standardized those to focus on big wins. I also replaced '0' with NaN as it was meaningless

TASK 4

Number Of Seats:

Before transformation the skewness was -0.35, indicating a left negative skew. The distribution shows a multimodal pattern with noticeable peaks around 19, 50, 150 and 190 seats. After Box-Cox transformation the skewness is closer to 0 and the distribution spreads more evenly across the range, though the peaks were preserved.

Capacity In Pounds:

Before transformation the skewness was 4.06, indicating a positive right skew. This is evident in the graph as the distribution shows the majority of aircrafts clustered at low capacity values with a very long right tail. After Box-Cox transformation the skewness is greatly reduced. From the graph, we see that the distribution becomes much more symmetric and bell-shaped.

TASK 5

Operating Status - Across all groups, the majority of aircraft are operating (Y), evident in the peaks across each size. Although the operating rates are all high, XLARGE aircrafts have the highest operating rate while SMALL aircraft have the lowest. This indicates that aircraft size has minimal impact on whether an aircraft is actively operating, and operating status may be conditional on other factors.

Aircraft Status - There is more distinction in groups. Status O (out of service) dominates all size groups and Status B is the second most common across all groups. In status A, LARGE aircraft have the highest proportion of actively operating aircraft, followed by XLARGE , while SMALL and MEDIUM have the lowest. Status L (leased) is invisible across all size groups, barely showing up on the graphs.

TASK 6

Model	Train RMSE	Test RMSE

LR - NUMBER_OF_SEATS	60.16	60.18
LR - CAPACITY_IN_POUNDS	87658.53	85438.28
RF - NUMBER_OF_SEATS	4.04	4.14
RF - CAPACITY_IN_POUNDS	51229.75	51126.15

For both Number Of Seats and Capacity In Pounds, Random Forest performs significantly better than Linear Regression as shown above in their RMSE Values. This suggests that a non-linear relationship exists and the near-identical train and test RMSE values shows that RF generalizes well to unseen data.